Multi-modal Safety Reasoning for Driving Actions

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Abstract

As autonomous vehicle research advances, the increasing complexity of deep neural networks often results in 'black box' models that lack transparency in decision-making. This project explores a multi-modal approach to provide action-based safety assessments backed by natural language reasoning. Our pipeline leverages a Faster R-CNN architecture for real-time object detection of road entities, the outputs of which are then processed by a CLIP-based vision-language model. By inputting both a driving scene and a proposed action (e.g 'move forward'), the system generates a binary safety decision supported by a human-readable justification. This dual-model framework is intended to enhance the safety and reliability of autonomous navigation systems.

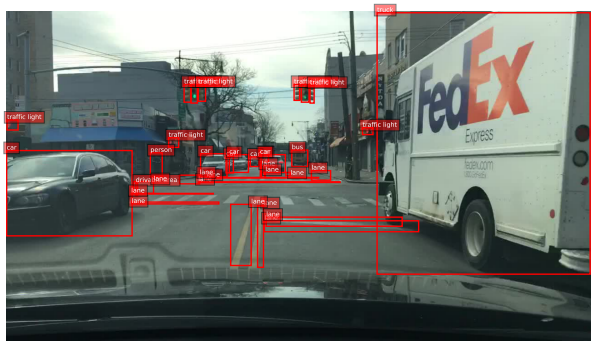

Figure 1. A sample image from the BDD100K dataset used in this work. The objective is to predict the red bounding boxes that identify road entities and drivable areas. Each image in the dataset is paired with annotations consisting of (x, y) coordinates and corresponding labels. We developed a script to visualize these bounding boxes. The dataset is publicly available.

1. Introduction

The maturation of Convolutional Neural Networks (CNNs) has transitioned image classification from a complex challenge to a foundational problem statement in Computer Vision. However, for autonomous vehicles, simple classification is insufficient; the system must not only be capable of identifying multiple road entities but also understand their relational context. To accomplish this, we propose a decoupled architecture. The primary phase utilizes a Faster R-CNN Object Detection architecture to identify objects and handle location. These spatial outputs then feed into a CLIP-based Vision Language Model (VLM) which outputs natural language reasoning and a safety score. In our work, we focus on driving scene images paired with object annotations obtained from a publicly available dataset (see sample images in Figure 1). In contrast to related work, we use another publicly available dataset of driving images paired with safe actions and human-verified justifications to train the second phase of the system.

2. Related Work

The ability to accurately localize and classify road entities is a key feature of autonomous navigation. Early benchmarks in the field were dominated by the transition from traditional gradient-based methods to Deep Convolutional Neural Networks. The introduction of the Faster R-CNN architecture by Ren et al. [2] surpassed these benchmarks by utilizing a Region Proposal Network (RPN) to share full-image feature convolutional features with the detection network, enabling near real-time performance.

Recent breakthroughs in multi-modal learning have bridged the gap between computer vision and natural language processing. The development of CLIP (Contrastive Language-Image Pre-training) by Radford et al. [1] demonstrated that models trained on massive pairings of images and text can learn semantic visual representations. While CLIP has been widely used for zero-shot classification, its application in generating safety justifications for driving maneuvers remains an emerging area of exploration.

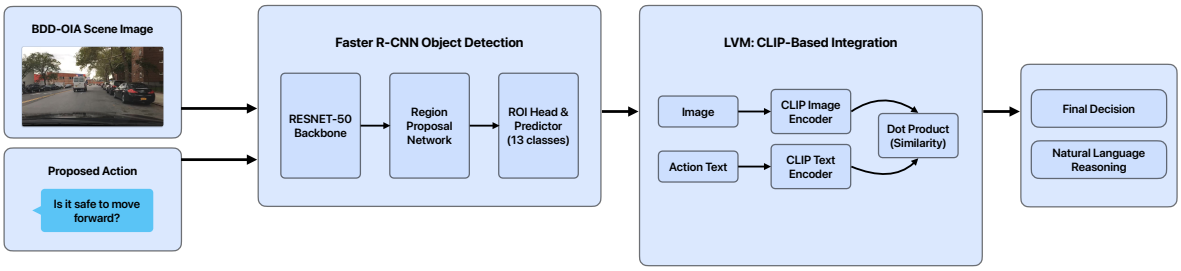

Figure 2. Here we show a figure detailing our model components. The input is a driving scene image and a proposed action. The output is a final decision indicating whether it is safe to perform the action backed by natural language reasoning.

Phase	Train-data	Epochs	Learning Rate	mAP@0.5	mAP@0.75	mAP@0.5:0.95
Faster R-CNN (RESNET-50)	BDD100K	5	5e-3	0.4571	0.2335	0.2513
Faster R-CNN (RESNET-50)	BDD100K	8	1e-4	XX	XX	XX

Table 1. Preliminary experimental results comparing the same architecture with different hyper parameters and image transformations. The second row uses an image transformation that horizontally flips 50% of all training images. XX marks pending results on planned experiments.

3. Methodology

The proposed system follows a modular, two-stage multi-modal pipeline designed to bridge the gap between raw perception and interpretable decision-making in autonomous driving. The architecture transitions from object detection to natural language reasoning in a sequential feed-forward manner.

3.1. Perception via Faster R-CNN

The first stage utilizes a Faster R-CNN framework to identify and localize critical road entities. A ResNet-50 backbone extracts high-level spatial features from the raw BDD100K input scene. A Region Proposal Network (RPN) then identifies potential objects of interest, which are then processed by a Region of Interest (ROI) Head and Predictor to classify the object in question. The predictor is configured to recognize 13 distinct classes (e.g., Car, Pedestrian, Traffic Light), ensuring the model captures the full semantic complexity of the BDD100k environment.

3.2. Reasoning via CLIP-based Integration

The second stage integrates visual detections with linguistic queries to evaluate action-based safety. Firstly, detected image regions from the previous stage and the Proposed Action text (e.g., "Move forward") are processed through parallel CLIP Image and Text Encoders. The system proceeds to calculate the Dot Product (Cosine Similarity) between the visual and textual embeddings to determine the alignment between the current scene and the proposed action. The pipeline concludes by generating a binary safety decision supported by natural language reasoning to justify the system's choice based on detected obstacles. [1]

4. Experimental Settings

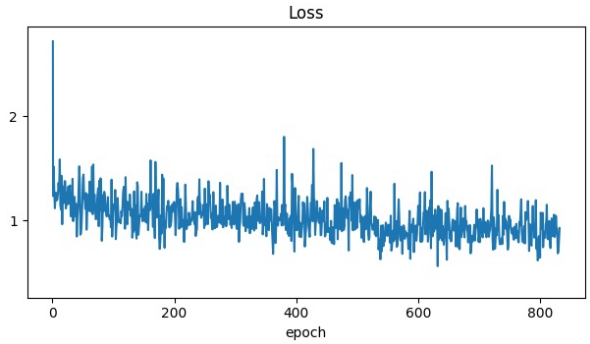

Figure 3. This figure shows the loss across 800 iterations (5 epochs). Each epoch is 175 iterations. A learning rate scheduler was introduced at around epoch 4 to stabilize the loss.
$Loss = L_{\text{obj}}^{\text{RPN}} + L_{\text{reg}}^{\text{RPN}} + L_{\text{cls}}^{\text{ROI}} + L_{\text{reg}}^{\text{ROI}}$

We used PyTorch to implement our system. The Faster R-CNN architecture is built using a ResNet-50 backbone and RPN network pre-trained on the COCO dataset. We replaced the ROI head with a new randomly initialized layer that was trained specifically on the BDD100K dataset. We chose not to freeze any of the pre-trained parameters so that the overall model is tuned to the BDD100K dataset, with the ROI head learning the most. This approach is known as Transfer Learning and was used to reduce both computational resources and training time. To measure how

[1] https://github.com/omar-dajani/safety-reasoning-driving-action

our model is performs, we calculated the loss, which is a multi-task objective function that forces the model to excel at localization and identification. The first two components originate from the RPN: the objectness loss ($L_{\text{obj}}^{\text{RPN}}$), which evaluates the model's ability to distinguish foreground objects and background scenery, and the RPN regression loss ($L_{\text{reg}}^{\text{RPN}}$) which measures the accuracy of the initial, rough bounding box proposals. The two remaining components are derived from the ROI head: the classification loss ($L_{\text{cls}}^{\text{ROI}}$) which penalizes the model for mislabeling categories (e.g. pedestrians), and the ROI regression loss ($L_{\text{reg}}^{\text{ROI}}$), which ensures the final bounding boxes are tightly fitted around to the object's boundaries. By aggregating these individual errors as shown in Figure 3, the model undergoes a backpropagation process that fine-tunes the entire architecture.

References

[1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pages 8748–8763. PMLR, 2021.

[2] S. Ren, K. He, R. Girshick, and J. Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 28, pages 91–99, 2015.